RELATIONS SATISFIED BY THE SHIFTED HIGHEST-ROOT VECTOR IN THE LEVEL-ONE DEMAZURE MODULE D(1,(K+1)THETA)

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Editorial note. All mathematical text is the author's own, copied from the fixed publisher source; the proof appears exactly as printed and nothing was rewritten.

Editorial note. The author's file declares no theorem environments of its own: they come from the SIGMA publisher class, which is not installed for this excerpt. The minimal preamble therefore reproduces that class's theorem-environment block (same environment names and styles); the author's own operator macros are taken from source lines 9-11.

Editorial note. Context blocks are the source material the proof uses: Garland's quoted identity, source lemma g, the definition of the level-one Demazure module, and the preceding source proposition WvsD (whose defining relations for D(1,(k+1)theta) are used at lines 489-490). They are marked as context and are not counted as this task's proof. Environment numbers printed here are local to the excerpt; the source statement is the lemma labelled crucial at source line 454 (Section 3).

Allowed standard prerequisites (editorial).

- Root systems and sl_2-triple arguments in a finite-dimensional complex simple Lie algebra. Roots, coroots, the highest root theta and d_alpha = 2/(alpha|alpha) are assumed. The weight computation for (x^-_theta (x) t^{2k+1-i}) wbar, the vanishing of its x^+_alpha (x) 1 image via theta - alpha in R^+, and the resulting sl_2 conclusion are retained obligations.
- Garland's current-algebra identity, as quoted by the author. The identity is quoted verbatim in the excerpt and is not re-proved. Its specialisation r = 2, s = 3k+1-i, alpha = theta and the subsequent manipulation of the resulting sum are the retained proof.
- Defining relations of the level-one Demazure module, obtained from the preceding source proposition. The excerpt carries the definition of D(1,lambda) and the author's presentation of D(1,k theta). The proof uses the resulting relations of D(1,(k+1)theta) at lines 489-490; nothing else about Demazure modules is assumed.

Lemma 1. *Given $s \in \mathbb{N}$, $r \in \mathbb{Z}_{\geq 0}$ and $\alpha \in R^+$, we have*

$$(x_\alpha^+ \otimes t)^{(s)}(x_\alpha^- \otimes 1)^{(s+r)} - (-1)^s \mathbf{x}_\alpha^-(r,s) \in \mathbf{U}(\mathfrak{g}[t])\mathfrak{n}^+[t] \bigoplus \mathbf{U}(\mathfrak{n}^-[t])\mathfrak{h}[t]_+.$$

We now specialize to the case $\lambda \in \mathbb{N}\theta$, and obtain some further useful relations that hold in $W(\lambda)$.

Lemma 2. *Let $k \in \mathbb{N}$. The following relations hold in the local Weyl module $W((k+1)\theta)$:*

1) $(x_\theta^- \otimes 1)^{2k+1}\big(x_\theta^- \otimes t^{2k+1-i}\big)w_{(k+1)\theta} = 0, \qquad \forall\, 0 \leq i \leq k;$
2) $(x_\theta^- \otimes t^m)(x_\theta^- \otimes t^{m+1})w_{(k+1)\theta} \in \big\langle \big(x_\theta^- \otimes t^{m+2}\big)w_{(k+1)\theta} \big\rangle, \qquad \forall\, m \geq k.$

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Definition 1. (see [4, Corollary 3.5]) The level one Demazure module $D(1,\lambda)$ is the graded quotient of $W(\lambda)$ by the submodule generated by the union of the following two sets:

$$\big\{(x_\alpha^-\otimes t^{s_\alpha})w_\lambda : \alpha\in R^+ \text{ such that } d_\alpha>1\big\}, \tag{1}$$

$$\big\{\big(x_\alpha^-\otimes t^{s_\alpha-1}\big)^2 w_\lambda : \alpha\in R^+ \text{ such that } d_\alpha=3 \text{ and } m_\alpha=1\big\}. \tag{2}$$

In particular, for $\mathfrak{g}$ simply laced, $D(1,\lambda)\cong_{\mathfrak{g}[t]} W(\lambda)$. We denote by $\overline{w}_\lambda$, the image of w_λ in $D(1,\lambda)$.

The following proposition gives explicit defining relations for $D(1,k\theta)$.

Proposition 1. *Given $k\geq 1$, the level 1 Demazure module $D(1,k\theta)$ is the graded $\mathfrak{g}[t]$-module generated by an element $\overline{w}_{k\theta}$, with the following defining relations:*

$$\mathfrak{n}^+[t]\,\overline{w}_{k\theta}=0,\qquad (h\otimes t^s)\overline{w}_{k\theta}=\langle k\theta,h\rangle\delta_{s,0}\overline{w}_{k\theta},\qquad s\geq 0,\quad h\in\mathfrak{h},$$
$$(x_\alpha^-\otimes 1)\overline{w}_{k\theta}=0,\qquad \alpha\in R^+,\quad (\theta|\alpha)=0,$$
$$(x_\alpha^-\otimes 1)^{kd_\alpha+1}\overline{w}_{k\theta}=0,\qquad \big(x_\alpha^-\otimes t^k\big)\overline{w}_{k\theta}=0,\qquad \alpha\in R^+,\quad (\theta|\alpha)=1,$$
$$(x_\theta^-\otimes 1)^{2k+1}\overline{w}_{k\theta}=0.$$

Lemma 3. *Let $k\geq 1$ and $0\leq i\leq k$. The following relations hold in the module $D(1,(k+1)\theta)$:*

1) $(x_\alpha^-\otimes 1)^{kd_\alpha+1}\big(x_\theta^-\otimes t^{2k+1-i}\big)\overline{w}_{(k+1)\theta}=0$, $\forall\,\alpha\in R^+$, $(\theta|\alpha)=1$;
2) $\big(x_\alpha^-\otimes t^k\big)(x_\theta^-\otimes t^{2k+1-i})\overline{w}_{(k+1)\theta}\in\big\langle\big(x_\theta^-\otimes t^{2k+2-i}\big)\overline{w}_{(k+1)\theta}\big\rangle$, $\forall\,\alpha\in R^+$, $(\theta|\alpha)=1$;
3) $\big(x_\theta^-\otimes t^{2k-i}\big)\big(x_\theta^-\otimes t^{2k+1-i}\big)\overline{w}_{(k+1)\theta}\in\big\langle\big(x_\theta^-\otimes t^{2k+2-i}\big)\overline{w}_{(k+1)\theta}\big\rangle$.

Proof. Let $\alpha\in R^+$with$(\theta|\alpha)=1$. This implies that $\theta-\alpha$ is also a root of $\mathfrak{g}$ and $(\theta|\theta-\alpha)=1$. We now prove part (1). Observe that, $(x_\theta^-\otimes t^{2k+1-i})\overline{w}_{(k+1)\theta}$ is an element of weight $k\theta$. Further $(x_\alpha^+\otimes 1)(x_\theta^-\otimes t^{2k+1-i})\overline{w}_{(k+1)\theta}=0$, since $(x_\alpha^+\otimes 1)\overline{w}_{(k+1)\theta}=0$ and $(x_{\theta-\alpha}^-\otimes t^{2k+1-i})\overline{w}_{(k+1)\theta}=0$, for all $0\leq i\leq k$. Considering the copy of $\mathfrak{sl}_2$ spanned by $x_\alpha^+\otimes 1$, $x_\alpha^-\otimes 1$, $\alpha^\vee\otimes 1$, we obtain part (1) by standard $\mathfrak{sl}_2$ arguments. We now prove part (2). Putting $r=2$, $s=(3k+1-i)$ and $\alpha=\theta$ in Lemma 1, we get

$$\big(x_\theta^-\otimes t^k\big)\big(x_\theta^-\otimes t^{2k+1-i}\big)\overline{w}_{(k+1)\theta}+\sum_{\substack{k+1\leq p\leq q\leq 2k-i\\ p+q=3k+1-i}}\frac{1}{(2\delta_{p,q})!}\big(x_\theta^-\otimes t^p\big)\big(x_\theta^-\otimes t^q\big)\overline{w}_{(k+1)\theta}$$
$$\in\big\langle\big(x_\theta^-\otimes t^{2k+2-i}\big)\overline{w}_{(k+1)\theta}\big\rangle, \tag{3}$$

since $(x_\theta^-\otimes 1)^{3k+3-i}\overline{w}_{(k+1)\theta}=0$, $\forall\,0\leq i\leq k$. Now we act on both sides of (3) by $x_{\theta-\alpha}^+$ and use the relation $(x_\alpha^-\otimes t^r)\overline{w}_{(k+1)\theta}=0$, for all $r\geq(k+1)$, which gives part (2). Part (3) is immediate from the part (2) of Lemma 2. □

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